1. The **Poisson** distribution is often used to describe how many times an event takes place within a certain time interval, such as number of daily customers at a store. Assume $x_1, \ldots, x_n \sim_{iid} Poisson(\lambda)$, where λ is the rate of occurrences. Calculate the maximum likelihood estimator of λ with pen and paper or LaTeX, showing your work. The probability mass function (PMF) for the Poisson distribution is:

$$f(x|\lambda) = \frac{\lambda^x e^{-\lambda}}{x!}$$

2. With `R`, generate data from $Poisson(\lambda = 8)$ with `rpois`, then use `optim` to estimate λ via the log-likelihood function for the Poisson distribution. Show your work. NOTE: You can take the log-likelihood function you found in the first question and convert that to something RStudio can understand.